NELVARX SYSTEMS AB
NELVARX MATHEMATICAL LABORATORY

GEOGRAPHY OF COMPUTE

Structural Architecture Assessment of AI Data Center Siting

PUBLIC RESEARCH DEMONSTRATION CASE

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Status	Synthetic structural demonstration

No real countries or sites are evaluated. No mathematical calculations. The internal manual is not disclosed.

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Companion document: Mathematical Modelling Research Report v0.2 (separate PDF).

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01 Executive Summary

The object is not a country or an actual site. It is a hypothetical large-scale AI data center architecture together with its critical external dependencies. The assessment concerns the structural coherence of the conditions required to create and operate it.

Central observation: cheap electricity, cold weather and available land do not individually establish a suitable configuration. Operational capability arises only when the relationships among compute, power, cooling, connectivity, control and mandatory permissions are coherent.

This report delivers a qualitative structural model, a dependency map, four synthetic cases, identified constraints and a list of evidence needed for further engineering evaluation. Equations, numerical calculations and optimization remain exclusively in the separate Mathematical Modelling Report v0.2.

Status: public demonstration based on synthetic conditions. No real sites were investigated. Actual construction feasibility has not been established.

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02 System Purpose and Scope

The mandatory purpose of the hypothetical system is to deliver a defined class of computing capacity with specified availability, connectivity, safety and controllability characteristics. Geographic placement is a means of achieving that purpose rather than the purpose itself.

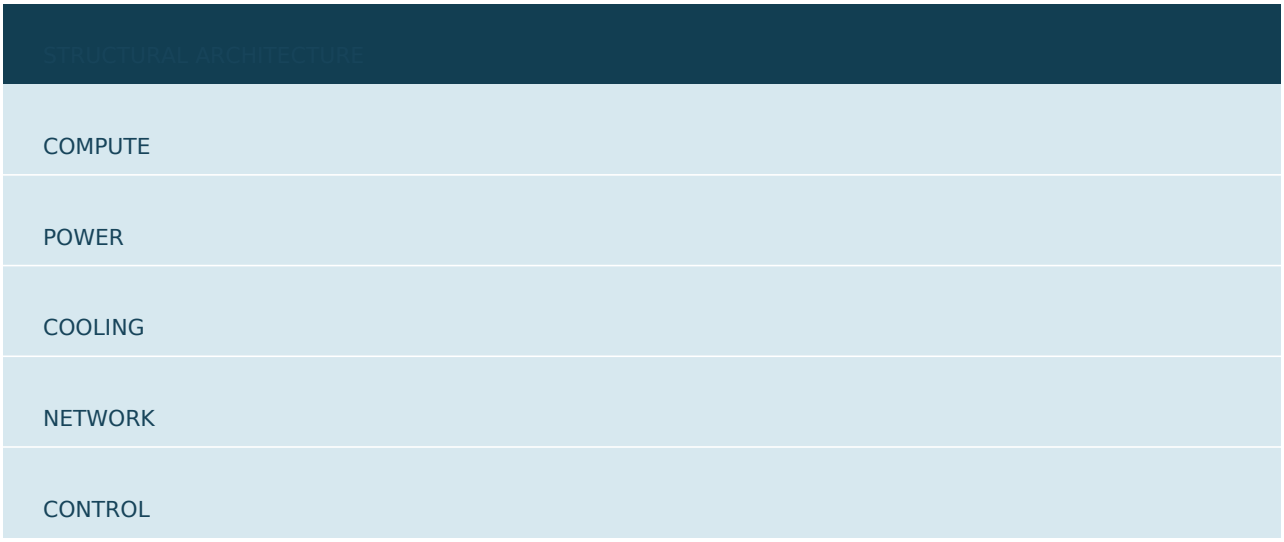
The internal architecture considered comprises compute racks, power distribution, cooling, networking, local operations, monitoring and safe-state mechanisms.

External dependencies include the power grid and utility operator, backbone networks, equipment suppliers, water or alternative heat rejection resources, permits, contractors and service providers. They may lie outside the operator’s direct control while remaining inside the analytical scope.

Outside this case: actual site surveys; detailed electrical or cooling design; formal legal confirmation; financial modelling; examination of actual countries or regions; mathematical computation; and operational testing.

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03 Structural Architecture



The compute domain receives workloads, operates processing equipment and storage, and generates electrical and thermal demand. The power domain provides supply, conversion and redundancy. The thermal domain removes waste heat. The network domain connects the facility to outside sources and recipients.

The control domain links telemetry, change authority, operating decisions, incident response and state records. None of these domains should be treated as autonomous: loss of an external resource may prevent the facility from meeting its purpose while the compute equipment itself remains functional.

Operational capability depends not on the mere existence of components but on admissible relations among them: supplied power must support total demand, cooling must support heat generation, network capacity must fit the workload, and operations must preserve controlled states.

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04 External Environment and Dependencies

Energy environment: access to connection capacity and reserves, grid development timing, consumption restrictions, dependence on the utility and applicable tariff conditions. A low average price does not establish available power.

Climate and environmental context: temperature and humidity regimes, seasonal variability, water limitations, heat rejection capability and local environmental impacts. A cold climate does not independently validate a cooling architecture.

Connectivity and supply: backbone connectivity, latency, path diversity, equipment availability, supplier dependencies, service support and maintainability.

Regulatory and organizational context: construction permissions, land-use status, data governance, reporting obligations, decision authority, contracts and third-party approvals. Physical possibility and regulatory permission are distinct.

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05 States and Transitions

Qualitative state sequence: considered configuration; verification of critical dependencies; design-aligned state; construction and connection; commissioning; normal operations; degraded mode; controlled stop or recovery.

A transition remains conditional until its required inputs, authorities, third-party obligations and completion criteria are established. Available land does not confirm readiness to construct; completed design work does not confirm that grid connection is available.

Entry into operation must preserve controllability, equipment protection and compliance with mandatory conditions. Recovery after an incident cannot be equated with restoration of electricity alone; cooling, networking and safe computing conditions also require confirmation.

The term “reachability” in this report is qualitative, describing a possible architectural route. Mathematical reachability definitions and timing analyses belong to the separate report.

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06 Constraints and Preserved Properties

Mandatory constraints include availability of necessary resources, technically and legally adequate connection, cooling compatibility with the workload, sufficient network connectivity, safety and regulatory compliance.

Properties intended to be preserved include avoidance of uncontrolled overload, observability of critical states, controlled shutdown, continuity of authority in degraded conditions and traceability of material operational changes.

A constraint differs from a preference. Lower electricity prices improve one comparison dimension but do not replace required access to capacity. Redundancy can be valuable, but independence of redundant resources must be established separately.

This report describes constraints verbally through structural relations, not equations. Numerical thresholds and their justification belong to mathematical and specialist engineering assessments.

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07 Failures, Degradation and Recovery

Loss of external power: a possible chain from utility failure to loss of load supply. Defined redundancy, shutdown capability and recovery conditions are necessary.

Insufficient heat rejection: functional compute components may become unusable when cooling fails. Power and thermal domains must therefore be assessed together.

Loss of external connectivity: some local computing activity may continue while the system fails to deliver its externally connected service purpose.

Loss of permission or essential resource access: technical readiness does not authorize operation. “Constructed” and “permitted and operationally ready” are different states.

Architectural recovery requires cause identification, verification of dependent subsystems, a recorded state transition and confirmed conditions for resuming operations.

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08 Four Synthetic Structural Cases

A (attractive energy conditions, uncertain connection). Cheap electricity and favorable climate do not remove uncertainty about a critical external dependency. Structural assessment: essential relation remains unverified.

B (aligned resources and specified operation). The synthetic description assumes coherence among power, cooling, network and control. Structural assessment: conceptually coherent configuration, not a verified real site.

C (resource conflict). Attractive energy conditions coexist with a hypothetical heat-rejection or cooling-resource constraint. Structural assessment: redesign or resolution of the conflict is necessary; favorable energy conditions do not remove it.

D (alternative with different trade-offs). Resources may be more expensive while dependency relationships and redundancy provisions are more clearly specified. Structural assessment: a distinct architectural alternative, not a demonstrated winner.

A–D are not countries, geographic regions or actual sites. These structural descriptions are qualitative and do not replace the separate mathematical classification.

A	Critical dependency unverified
B	Conceptually coherent, not verified
C	Structural conflict
D	Alternative architecture

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09 Structural Findings and Risks

Finding 1. Resource attractiveness does not establish architectural coherence. Finding 2. External dependencies remain analytically relevant even when outside the operator’s organizational boundaries.

Finding 3. Workload purpose must precede geographic selection. Finding 4. Critical relations require site-specific evidence; until then, the assessment remains preliminary.

Principal risks: unconfirmed grid capacity; false independence of redundant resources; inadequate heat rejection in extreme conditions; unknown connectivity capability; regulatory limitations; and loss of control during incidents.

These risks are possible modes of structural degradation, not recorded incidents at real facilities. Their empirical likelihood and financial severity are not calculated here.

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10 Result Boundaries and Conclusion

This public structural assessment demonstrates how a potential computing facility connects resources, functions, constraints and external decisions into a single architectural system. It does not prove that any real site can be built.

Central result: before ranking locations by cost or climate, investigators must establish the compatibility of mandatory components and external dependencies, define relevant states and transitions, identify structural contradictions and specify required evidence.

For a real project, the next step would require workload requirements, verified utility and resource conditions, permitting information, interface documentation and reviews by qualified domain specialists. Only then can a site-specific conclusion be formulated.

Disclosure boundary: this document presents qualitative structural results but does not disclose Nelvarx Systems AB’s internal production protocol or proprietary know-how. A separate Mathematical Modelling Report v0.2 presents a formal research model, synthetic calculations and parameter tests.

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